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Faculty of Electrical Engineering

Department of Computers & Electrical Engineering

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Term paper on the subject of

Metasurface optical elements based on the Pancharatnam-Berry geometric phase

Review of the scientific journal article

D. Lin, P. Fan, E. Hasman, and M. L. Brongersma, “Dielectric gradient metasurface optical elements,” *Science*, vol. 345, no. 6194, pp. 298–302, Jul. 2014.

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Abstract

Metasurface materials are used to replace bulky optical components with flat optical elements, but existing metallic metasurfaces suffer from low efficiency in transmission mode. In the research paper, the authors report the experimental realization of dielectric gradient metasurface optical elements, based on silicon nanobeams, that achieve high efficiency in transmission mode for the visible spectrum. The components utilize the geometric Pancharatnam-Berry phase for full wavefront control; ultrathin (100-nm) gratings, lenses, and axicons were demonstrated. The fabricated devices showed high performance, including the generation of a non-diffracting Bessel beam from an axicon, a beam-deflector diffraction efficiency of 75% that was measured at a 500-nm wavelength, and metalens focusing to a near-diffraction-limited spot. This development paves the way for low-cost, lightweight, and compact optical components that can be easily integrated with electronics using mature semiconductor fabrication technologies.

Keywords: metasurfaces, silicon nanophotonics, dielectric gradient, geometric phase, high diffraction efficiency, axicon (Bessel beam), metalens, beam deflector, half-wave plate.

1. Introduction

When we hear the term “optical component”, the first image that comes to mind is a bulky conventional glass lens much like in space telescopes or sports cameras. An optical element of micrometer scale and nanometric thickness might sound like science fiction, but it's a reality realized by the innovative technology of meta-surfaces - artificial thin materials composed of nano-structures named “meta-atoms”. The paper's fundamental building block, or meta-atom, is a silicon (Si) nano-beam antenna with a thickness of just 100-nm, at least a hundred times thinner than a typical strand of human hair ($>10\text{ }\mu\text{m}$ thick).

Carefully designing the dimensions, materials, geometry and orientation of meta-atoms allows for accurate manipulation of an incident light beam's wavefront. Since diffraction is dictated by the phase of the light's electromagnetic fields along the aperture surface, mimicking the phase response of a desired optical device will result in a similar macroscopic optical behavior by a meta-surface. Theoretically, this opens up the possibility of realizing any optical device as a meta-surface, allowing for the minimization of devices to lightweight components in the scale of mere micrometers.

The first attempts to create such flat optical elements were based on metallic nano-antenna arrays. While these were effective in manipulating light in reflection mode, they suffered from high ohmic losses (heat), which resulted in very low efficiencies when operating in transmission mode. This fundamental limitation prevented their use in applications requiring light to pass through the device, such as lenses.

2. Dielectric gradient devices

In their research paper "Dielectric gradient metasurface optical elements" (DGMOEs), the researchers Lin, Fan, Hasman and Brongersma have demonstrated an experimental application of Pancharatnam-Berry phase Optical Elements (PBOEs) that achieves high working efficiencies for the visible light spectrum in transmission mode. To overcome the limitations of ohmic devices, the authors prepared a thin (thickness of 100 nm) layer of poly-Si nano-beams deposited on a SiO_2 quartz substrate and generated in a single patterning step.

These nano-beams act as resonant optical nano-antennas, with the intended behavior of a half-wave plate - an optical element that introduces a 180-degree phase shift between

orthogonal polarization axes, thereby reversing the handedness of circularly polarized light.

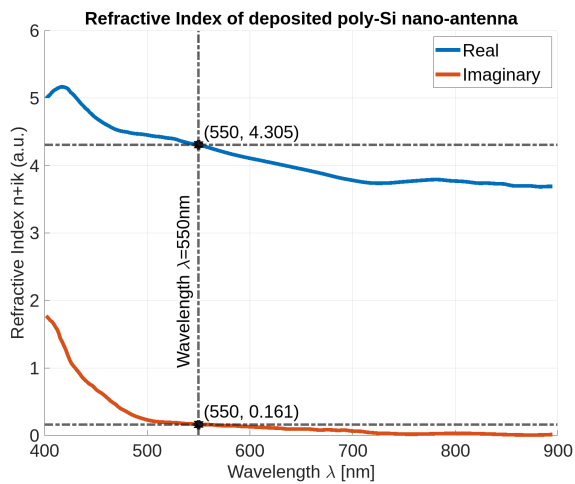


Fig. 1. Refractive index of the poly-silicon (poly-Si) films making up the DGMOEs, real (blue) and imaginary (red) parts. Extracted from the Fig. S1A. of the supplementary materials via a plot digitization program. At $\lambda = 550 \text{ nm}$ we get $\sim 4.305 + 0.161i$. The imaginary part of the refractive index directly affects the absorption coefficient.

The researchers realized a 120-nm wide and 100-nm high Si nanobeam, with a duty-cycle of 60% that corresponds to a nano-beam period of 200-nm. Then, the researchers have illuminated the structure with linearly polarized light, with perpendicular polarization to the nano-beam (x-axis) and parallel to it (y-axis). They have conducted various simulations, measuring the absorption, reflectance, phase, and transmittance of the structure, over a span of different wavelengths (400-nm to 1200-nm).

In the research paper, the terms used are Transverse Electric or Magnetic (TE/TM) illuminations, however this terminology is inaccurate in the way seen in Fig. 2. here. For this reason, we will use x-polarization instead of TE, and y-polarization instead of TM, which are more accurate terms for our purposes. We also have to take note that the researchers have defined the x-axis to be parallel to the nano-beam's width, and the y-axis to be parallel to its length.

To allow TE and TM polarizations to describe the response to x-polarization and y-polarization, these terms should describe the polarization with respect to the xz-plane.

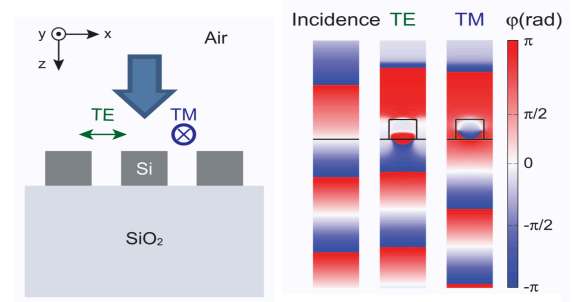


Fig. 2. A schematic view of the half-wave plate under illumination (left), and the wavefront phase response (right). Fig. 2A. & Fig. 2B. from the research paper. An important thing to note is the definition of the axes in the leftmost figure. The wavefront response is given for no waveplate (left), and with a waveplate present: for x-polarized light (middle) and y-polarized light (right).

At a wavelength of 550-nm, this resulted in a phase retardation of $\sim 0.14\pi$ radians for the x-axis and $\sim 1.15\pi$ radians for the y-axis, both relative to the incident wave's phase. This is a total relative phase shift between the two polarization axes of $\sim 1.01\pi$ radians, which is in $\sim 1\%$ error relative to the behavior of a half-wave plate. The researchers have also provided an experimental measurement of the phase retardation, which agrees with the paper's stated results (as seen in Fig. 3. in this document).

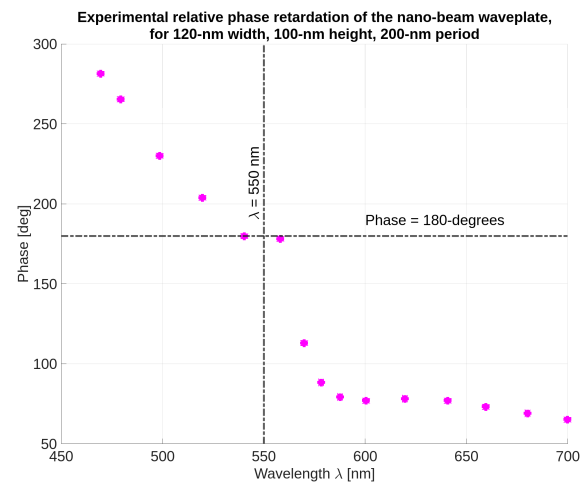


Fig. 3. Experimental relative phase retardation (between the polarization axes) extracted from the Fig. S5E. of the supplementary materials via a plot digitization program. For $\lambda = 550 \text{ nm}$ we get $\phi \approx \pi [\text{rad}]$, although the exact wavelength of 550-nm was not explicitly sampled.

The black line in the wavefront response seen in Fig. 2B. is the point along the propagation axis where the half-wave plate is present,

which causes a 180-degrees phase shift within the black square (symbolizing the nano-beam). This phase shift causes the entire phase profile to be shifted in space, as we can see by comparing the results for an incident wave with no waveplate present.

The transmission coefficients of a half-wave plate should be equal for both axes. However, the experimental transmission coefficients vary significantly at wavelength of 550-nm, with $t_x \approx 0.175$ and $t_y \approx 0.5$. This result was attributed by the researchers to imperfections in the fabrication of the nano-structures, or the birefringent nature of the nano-beams being size-dependent. However, when looking at the simulation data provided by the authors (Fig. S5C. in the supplementary materials of the paper), it seems that even theoretically speaking, there is no evidence that this structure should have equal transmission coefficients $t_x \approx t_y$ at a wavelength of 550-nm.

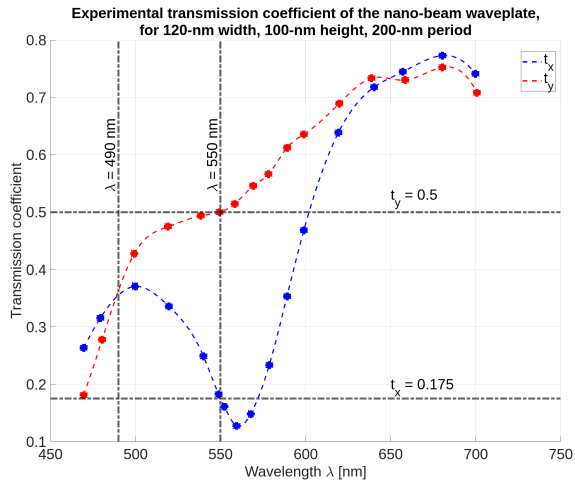


Fig. 4. Experimental transmission coefficients t_x (blue) and t_y (red) extracted from the Fig. S5C. of the supplementary materials via a plot digitization program. For $\lambda = 550 \text{ nm}$ we get $t_x \approx 0.175$ and $t_y \approx 0.5$. We can also notice that an interception is expected at around $\lambda \approx 490 \text{ nm}$.

The simulation graphs seem to intersect only around the $\lambda \approx 500 \text{ nm}$ point, which is in close similarity with the experimental data provided in Fig. S5C (as seen in Fig. 4. in this document). Later we will see that this discrepancy has negative consequences on the results for the beam-deflector DGMOE.

According to Fig. S5A. in the supplementary materials and Fig. 2D. in the scientific journal article, the absorption spectra of the nano-beam waveplate seems to be equal at a wavelength of 550-nm. However, this is a simulated result and the authors have not provided experimental data on the matter. Moreover, when interpolating the experimental data points for the transmission coefficients t_x , t_y and the axial relative phase φ , and calculating the Cross-Polarization Efficiency (CoPE) and Co-Polarization Efficiency (CrPE) defined by:

$$\begin{aligned} \text{CoPE} &= \left| \frac{1}{2} (t_x + t_y e^{i\varphi}) \right|^2 \\ \text{CrPE} &= \left| \frac{1}{2} (t_x - t_y e^{i\varphi}) \right|^2 \end{aligned} \quad (1)$$

we get that the maximum CoPE and minimum CrPE are actually achieved at a wavelength of approximately $\sim 520 \text{ nm}$. It's important to note that at such a wavelength, the axial relative phase shift differs significantly from a mere 180-degrees shift.

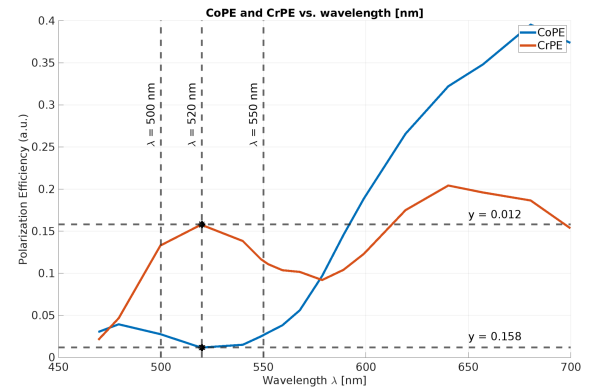


Fig. 5. Result of interpolating the transmission coefficients t_x and t_y , and the relative phase shift φ , to perform a numerical calculation of the CoPE and CrPE. It's noteworthy that the extremum points are locally (near 550-nm) achieved at $\sim 520 \text{ nm}$ wavelength.

Thus the guiding principle for the design of the Si nano-antenna seems to have been simply achieving a phase retardation as close as possible to that of a half-wave plate (180-degrees), while maintaining almost equal absorption fractions for both polarization axes. Considering the researchers had the intention to overcome the ohmic losses of metallic

metasurfaces, it's fair to assume that this was one of the guiding principles in the Si nano-antenna design.

Moreover, the authors claim that “the presented design principles are generally applicable to any high-index semiconductor or insulator and can be applied at any desired wavelength”. According to the experimental and simulation results, doing so will require choosing different materials for the nano-antenna and substrate.

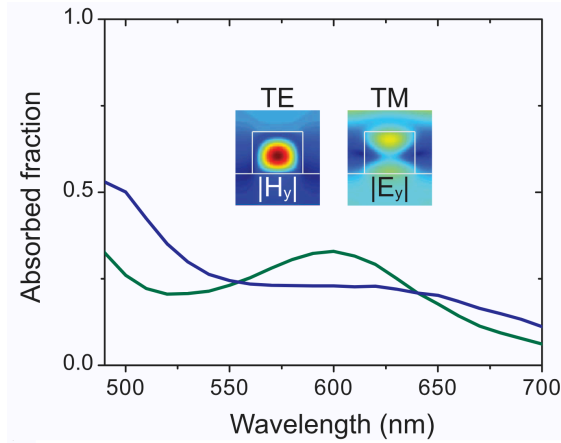


Fig. 6. Simulated absorption coefficients for x-polarized light (green) and y-polarized light (blue), from Fig. 2D. of the paper. For $\lambda \approx 550 \text{ nm}$ we get $t_x \approx t_y$, which indicates equal energy losses in the form of heat for both axes. Taking the TE and TM polarizations with respect to the xz-plane makes the small field figures make sense.

The authors of the paper then decide on a grating period of $\Lambda = 3.2 \mu\text{m}$ for all the DGMOEs in the paper, using eight phase discretization levels. The nano-beam shape is identical for both an orientation of 0-degrees and 180-degrees, and thus the phase profile relates to the orientation profile:

$$\phi(x, y) = 2\theta(x, y) \quad (2)$$

where $\phi(x, y)$ is the phase profile of the DGMOE, and $\theta(x, y)$ the spatial orientation profile. The discretization into eight phase levels results in the orientation profile being in increments of 22.5-degrees, or $\pi/8$ radians, resulting in phase increments of 45-degrees, or $\pi/4$ radians.

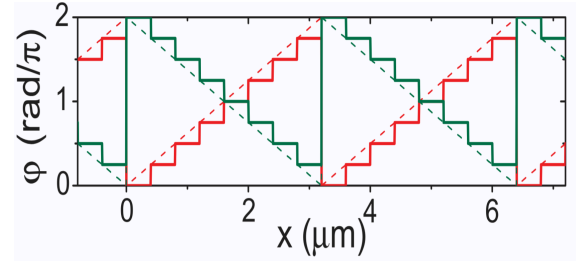


Fig. 7. Phase profile of a blazed grating (used as a beam deflector in the journal article) for incident LHCP light (red) and RHCP light (green). The dashed lines are the continuous phase profile that was discretized by the solid lines. For every four steps we get an increase of $1 \text{ rad}/\pi$, or in other words an increment of $\pi/4$ radians per step.

3. Bessel beam (axicon)

Bessel beams are a class of beams prized for their remarkable non-diffracting and self-healing properties. Unlike conventional Gaussian beams that spread out as they travel, a Bessel beam maintains a tightly focused central spot over a significant distance. If this central spot is blocked by an obstacle, the beam can reconstruct itself on the other side. These unique characteristics make Bessel beams invaluable for a wide range of applications, including high-resolution microscopy and precision lasers.

For this reason, the researchers chose to start their paper with this result as it is a strong selling point - demonstrating the ability to create a complex optical device (axicon) with nanometric thickness, that achieves a result with immediate real life applications (the Bessel beam).

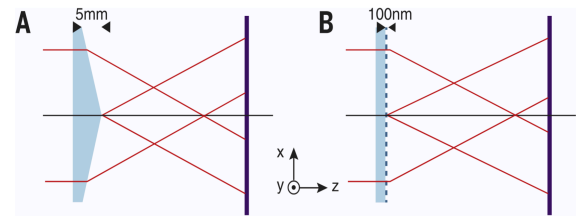


Fig. 8. A schematic focused on achieving the same optical behavior as a classical axicon with millimeter-scale thickness (A), using a nanometric-scale device (B). (research paper's Fig. 1A. and Fig. 1B.).

To recreate the xz-plane cross-section intensity profile provided in Fig. 1E. in the paper, I ran a simulation of the

Huygens-Fresnel integral for an incident Gaussian beam. The simulation took into account a metasurface of radius $R = 32 \mu\text{m}$ (according to section 9 in the supplementary material) with a two-dimensional linear (with respect to $r = \sqrt{x^2 + y^2}$) phase profile:

$$\phi(r) = 2\pi r/\Lambda \quad (3)$$

which can be inferred from Fig. S8. of the supplementary materials. The electric field at the metasurface plane given an incident Gaussian beam:

$$E(r, z = 0) = t_A(r) e^{-r^2/w^2} e^{i\phi(r)} \quad (4)$$

where $w = R/2$ is the Gaussian beam's width, and $t_A(r) = \text{circ}(r/R)$ is the aperture transmission function of the metasurface. The Gaussian beam width is unknown from the data in the paper, but for simulation purposes it's best to take a wide beam.

My simulation was conducted for a wavelength of $\lambda = 550 \text{ nm}$, within a $15\text{-}\mu\text{m}$ radius at the observation xy-plane, and within a $40\text{-}\mu\text{m}$ radius at the source xy-plane, $200\text{-}\mu\text{m}$ along the optical z-axis. The amount of sample points per axis was 400×400 for the source xy-plane at the source ($z = 0$), and 300×2000 for the observation xz-plane. This results in a resolution of 100-nm ($\sim \lambda/5$) for every axis.

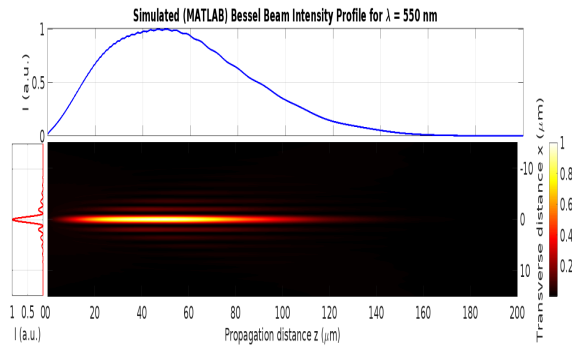


Fig. 9. Recreation via MATLAB simulation of the Bessel beam intensity profile at xz-plane for $\lambda = 550 \text{ nm}$ (Fig. 1E. from paper). The image resolution is 300×2000 .

As we can see by measuring the ratio between the central beam intensity and the first outer-ring's intensity, it is approximately 6:1, indicating that the resulting beam in the simulation is indeed a Bessel beam. In the supplementary materials, there is an additional experimental result for a Bessel beam generated by an incident Gaussian beam at wavelength of 700-nm .

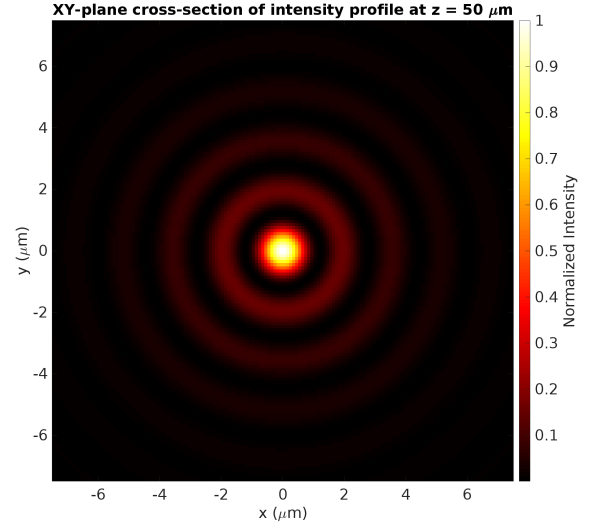


Fig. 10. MATLAB simulation of the Bessel beam intensity profile at the xy-plane for $\lambda = 550 \text{ nm}$ (recreation of Fig. 1C. from the journal article). Image resolution is 150×150 , since this colormap was plotted in a $7.5\text{-}\mu\text{m}$ radius to enhance visibility. Spot-size is $\sim 1.2\text{-}\mu\text{m}$ for the inner dot.

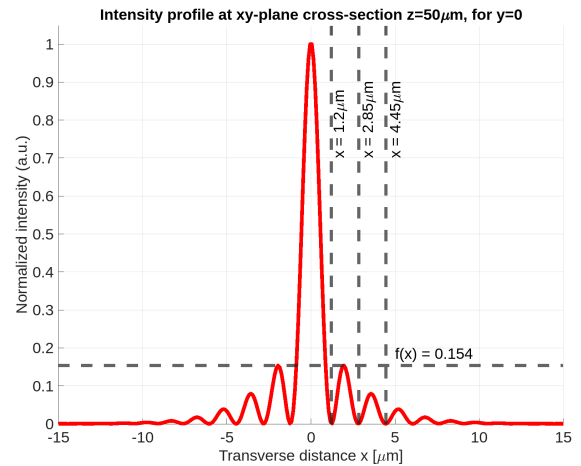


Fig. 11. Intensity profile cross-section. We can see that the ratio between the central beam and the first ring is around 0.154 (approximately 6:1), a signature of a Bessel beam.

The simulated beam exhibits the 6:1 intensity ratio between the innermost spot and first ring, and has a maximum intensity at around $\sim 50\text{-}\mu\text{m}$

from the metasurface along the optical axis. Furthermore, it has a spot-size radius of $\sim 1.2\text{-}\mu\text{m}$ for the inner spot, a radius of $\sim 1.65\text{-}\mu\text{m}$ for the first and second rings. Numerically integrating the intensity over the main spot, and the first and second rings, we get that the total intensity of the inner spot is $I_0 \approx 529 \cdot 10^{-9}$ and the total for both the first and second rings is $I_1 \approx 122 \cdot 10^{-9}$ and $I_2 \approx 64.9 \cdot 10^{-9}$. This implies that most of the beam's energy is concentrated in the center spot, with a rapidly falling off intensity concentration as we move further from the center.

The results closely resemble those in the paper. The main thing to note is that the base angle of the axicon DGMOE is given (in section 9 of the supplementary materials) by $\beta = \lambda/\Lambda$. This means that the base angle should vary for a 700-nm incident beam compared with a 550-nm beam. In the case of a 550-nm incident beam, the base angle is $\beta \approx 0.172 \text{ rad}$, or ~ 9.86 -degrees.

4. Beam deflection

A blazed grating is a fundamental optical device engineered to function as a highly efficient beam deflector. Unlike classical gratings, which rely on a sawtooth height profile, the DGMOE relies on imitating their phase profile to redirect nearly all of the incident light into a single, desired direction. This ability to efficiently steer light into a precise direction makes beam deflectors a cornerstone component with various real world applications such as in spectrometry, LIDAR scanning, and integrated photonic circuits. The DGMOE's phase profile is a one-dimensional linear profile given by the equation:

$$\phi(x) = 2\pi x/\Lambda \quad (5)$$

An ideal beam deflector is dictated by the polarization efficiencies (CoPE and CrPE), and the incident field's polarization state's inner

product with the circular polarization vectors acting as a basis for the Stokes parameters vector space. The output wave E_{out} will have a polarization state with the relation:

$$|E_{out}\rangle = \sqrt{\eta_E}|E_{in}\rangle + \sqrt{\eta_R}e^{i2\theta(x,y)}|R\rangle + \sqrt{\eta_L}e^{-i2\theta(x,y)}|L\rangle \quad (6)$$

where the efficiencies are given by:

$$\begin{aligned} \eta_E &= \left| \frac{1}{2}(t_x + t_y e^{i\varphi}) \right|^2 \\ \eta_R &= \left| \frac{1}{2}(t_x - t_y e^{i\varphi}) \right|^2 < L|E_{in}\rangle^2 \\ \eta_L &= \left| \frac{1}{2}(t_x - t_y e^{i\varphi}) \right|^2 < R|E_{in}\rangle^2 \end{aligned} \quad (7)$$

using the bra-ket notation denoting an inner product or a basis vector. This relation implies that the amount of polarization from the incident wave that is conserved is directly correlated to the CoPE. In the case of an ideal half-wave plate, $t_x = t_y$ ($= 1$) and $\varphi = \pi [\text{rad}]$, which causes $\eta_E \approx 0$, meaning we achieve complete beam-deflection.

As we discussed previously, the researchers have not achieved the uniform behavior of a half-wave plate, but instead have $t_x \neq t_y$. This explains why their experimental results exhibit poor beam deflection. For example, at a wavelength of 550-nm we've established that $t_x \approx 0.175$, $t_y \approx 0.5$ and $\varphi \approx \pi [\text{rad}]$. This results in a co-polarization efficiency of $\sqrt{\eta_E} \approx 0.165$ and cross-polarization efficiency of $\sqrt{\eta_R} \approx 0.335$.

These efficiencies are an indication of the polarization state of the electric field, meaning the ratio between the intensity of the deflected and non-deflected spots should be η_R/η_E , or approximately a 4.1:1 ratio. This means only $\sim 80\%$ of the beam that is successfully transmitted is being deflected, with $\sim 20\%$ leaking to the undeflected path.

Note that this is relative to the transmitted wave, and it gets worse when we take into

account the losses in the meta-surface: from a quick calculation, one can see that for a wavelength of 550-nm, only ~14% of light actually passes through the device, a result from summing $\eta_E + \eta_R \approx 0.14$. The other ~86% are lost due to reflection (>50% according to Fig. S5B. in the supplementary materials) and absorption (~25% according to Fig. 4. absorption coefficient in this document).

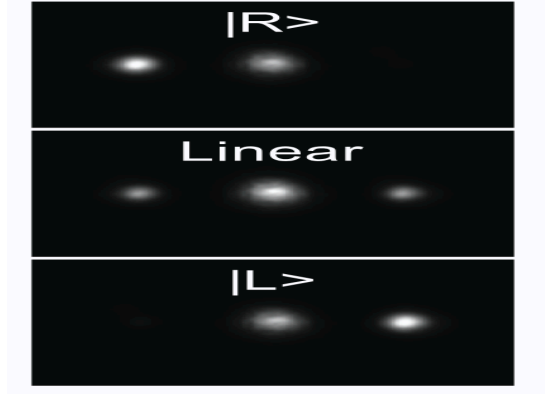


Fig. 12. Measured diffraction patterns under different illumination polarizations (from paper's Fig. 3C.). For every incident polarization state, the non-deflected beam is located in the middle, and the other spots are deflected beams. This is validated by the fact that for circular polarization, the brightest spot is not in the middle.

The diffraction efficiency of the device is mainly affected by the first-order diffraction, according to the researchers, and can be estimated by $\eta_D \approx I_{1st}/(I_{1st} + I_{0rd})$. In other words, it is the ratio of the deflected power and the total transmitted power. Further idealizing their model, we can simulate the diffraction efficiency in MATLAB by taking $\eta_D \approx \eta_R/(\eta_R + \eta_E)$ (for a LHCP incident wave), resulting in a plot that closely resembles the one in Fig. 3D. from the paper. The researchers attributed the fact that the efficiencies are experimentally lower to fabrication imperfections and assumed that the birefringent nature of the nanobeam is size-dependent (which was not taken into account in their simulations).

The last key result for the DGMOE relates to the steering angle of the beam deflector. The steering angle α determines the angle of deflection of the incident beam, and obeys the

formula $\sin(\alpha) = \lambda/\Lambda$. This means α increases for longer wavelengths, which is confirmed by Fig. 1E. from the paper.

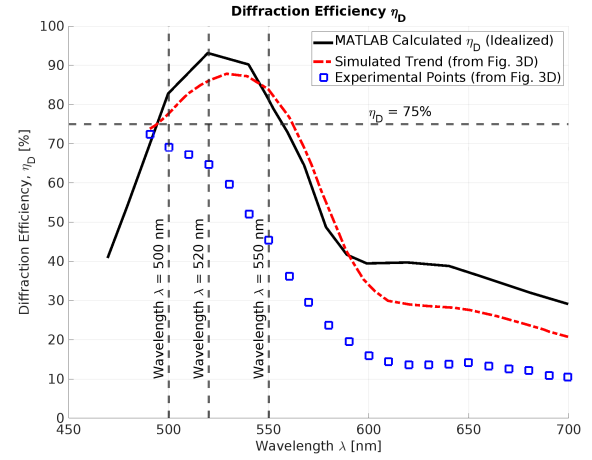


Fig. 13. Diffraction Efficiency η_D . Black - calculated using the idealized model, red and blue - extracted from Fig. 3D. using a plot digitization program. We notice that experimentally, the diffraction efficiency hasn't reached 75% for 550-nm wavelength like the paper claims. The idealized model closely resembles the model the researchers used to calculate η_D .

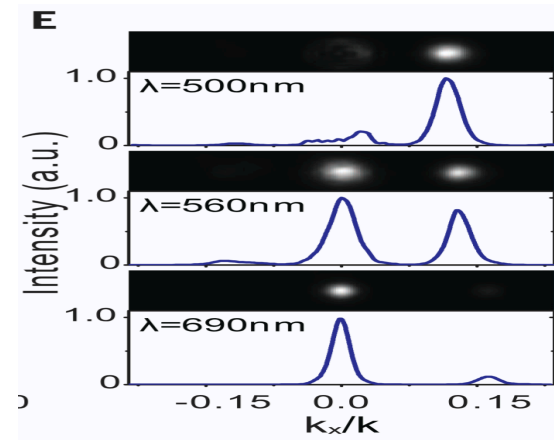


Fig. 14. Deflection pattern for incident LHCP light at varying wavelengths (from paper's Fig. 1E.). As the wavelength increases, the steering angle increases, causing the dot to stray further away from the center (non-deflected) spot. Due to the diffraction efficiency dropping as the wavelength increases, we get that for 500-nm most of the beam gets deflected, for 560-nm we get poor deflection (but not zero), and for 690-nm we get close to no deflection (the furthest distance from the center yet).

5. Metalens

Lenses are arguably the most essential optical element there is, allowing millions of people worldwide to have clear eye vision, and enabling the function of cameras. The ongoing

drive to create lightweight low-cost lenses is key for the development of the field of optics, and the shrinking of their thickness to nanometric scale allows easier application of compact elements such as augmented reality devices (such as those in the helmets of pilots).

The researchers have created a DGMOE that acts as a conventional lens by imitating the hyperboloidal phase profile of a lens:

$$\phi(r) = 40\pi + \frac{2\pi}{\lambda} \left(f - \sqrt{r^2 + f^2} \right) \quad (8)$$

where f is the focal length of the metalens, and $r^2 = x^2 + y^2$. The phase profile has an initial phase of $\phi(r = 0) = 40\pi$, which doesn't really affect the phase profile since it's divisible by 2π , and even if we ignore this term we will get the same results.

For a wavelength of 550-nm, the researchers mention that the metalens features a focal length of 100- μm , resulting in a Numerical Aperture (NA) of 0.43. This indicates that the fabricated lens have an aperture diameter D of 96- μm , since the NA is:

$$NA = n \cdot \sin(\theta) \quad (9)$$

where n is the refraction index of the medium the lens is positioned within, and $\tan(\theta) = D/2f$.

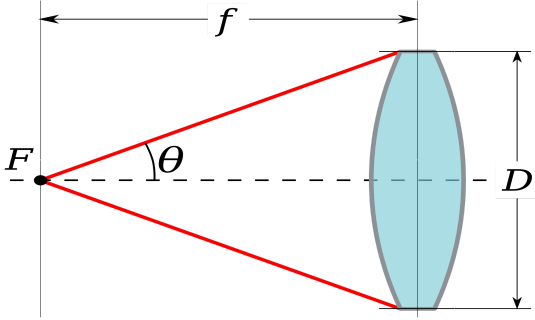


Fig. 15. A conventional lens with focal length f and aperture diameter of D . The angle θ is used for the calculation of the lens's NA.

The Fresnel Number (FN) of the metalens is:

$$N_F = (D/2)^2 / (\lambda f) \quad (10)$$

giving $N_F \approx 41.89$, a very decent FN value, indicating the lens is operating in the near-field

diffraction regime, obeying the geometric optics approximation closely. In other words, the focal-length f is a well-defined property of the metalens.

One thing to note is that the NA is classically independent of the wavelength, however the meta-surface exhibits a wavelength-dependent behavior (such as the relative phase retardation varying with λ). This behavior should affect the focal length, and thus the NA will depend on λ .

To recreate these results, I ran a MATLAB simulation for $\lambda = 550 \text{ nm}$ and $f = 100 \mu\text{m}$ with the hyperboloidal phase profile, sweeping the source xy coordinates at a 48- μm radius, and the observation plane at 10- μm radius, observing on the optical axis at $\pm 25 \mu\text{m}$ around the theoretical focal length (100- μm). The resolution for all axes was taken to be 0.1- μm ($\sim \lambda/5$), with a source plane resolution of 1000x1000, an observation xy-plane resolution of 200x200, and xz-plane resolution of 200x500. The MATLAB results closely resemble those in the paper, with a focal-shift of $\sim 0.05 \mu\text{m}$, and spot size (FWHM) of $\sim 0.6 \mu\text{m}$ diameter.

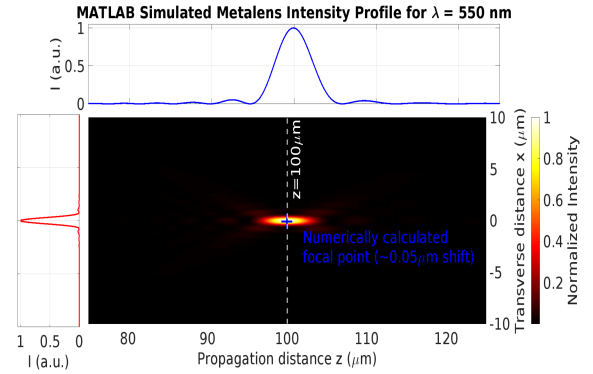


Fig. 16. Intensity profile for xz-plane observation (recreation of Fig. 1C. from the paper). The focal point experiences little to no shift, great results for this DGMOE. Image resolution is 200x500.

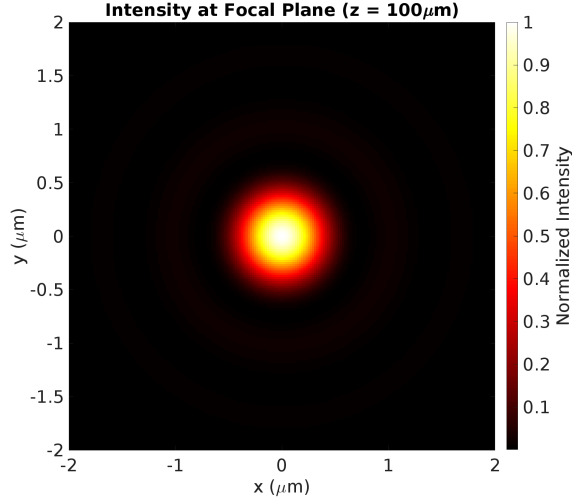


Fig. 17. MATLAB simulation of the intensity profile at the focal length ($z = f$) at the xy-plane for $\lambda = 550 \text{ nm}$ (recreation of Fig. 1C. from the journal article). The resolution is 200×200 . Spot-size (FWHM) is $\sim 0.6 \text{ }\mu\text{m}$.

6. Numerical DGMOEs results

Simulations of the nano-beam and of the DGMOEs have been performed in Ansys's Lumerical numerical program. First, we conducted a sweep on the meta-atom design parameters, the width and period of the nano-beam, with widths varying from 80-nm to 160-nm and periods of 160-nm to 240-nm, with a 2-nm step increment. This results in a 41×41 grid, at 550-nm wavelength, for both simulations with x-polarized and y-polarized incident light.

The simulation results for the meta-atom reveal that the optimal design point, where we achieve minimal CoPE, is for 136-nm width and 230-nm period, or $\sim 59.1\%$ duty-cycle. For these dimensions we achieve a phase difference with $<1\%$ error from π , and a transmission coefficient ratio of $t_x/t_y \approx 1$. This means we achieve an accurate behavior of a half-wave plate, in contrast to the design chosen in the research paper.

However, we pay a price for this improvement - the CrPE of the paper's design is around $\sim 12\%$, while this new design has a CrPE of $\sim 4.4\%$. Simply maximizing the CrPE does not achieve the desired half-wave plate behavior - as seen in Fig. 18. below, we can see that the

maximal CrPE ($\sim 19.1\%$), with a phase difference of $\sim 1.01\pi$ and $t_x/t_y \approx 0.55$.

Moreover, the maximal CrPE point has a CoPE of $\sim 1.6\%$, while the paper's CoPE is $\sim 2.2\%$; in our chosen design point, the CoPE is $<0.01\%$.

If we simply take the goal of minimizing the device losses as a guiding principle, it seems that a 160-nm width and 174-nm period achieves this behavior. But the CrPE alone is not a sufficient performance metric, since we might experience unwanted behavior due to the $\sim 1.6\%$ CoPE (as seen in the beam-deflector chapter).

Another key result to note is that the phase we received for the paper design is $\sim 0.973\pi$, compared to the paper's $\sim 1.01\pi$. This can be explained by the fact that our simulations were conducted in Lumerical, compared to the paper's COMSOL simulations. Moreover, the plots for the material dielectric real and imaginary parts were extracted via a plot digitizer from the paper's graphs, which could introduce a deviation from the actual material properties, and thus a deviation in ϕ .

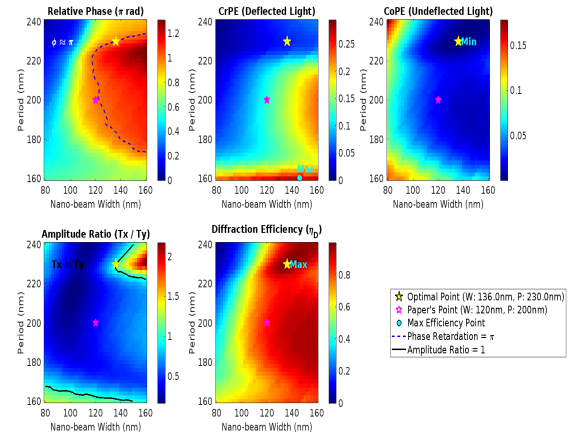


Fig. 18. Numerical simulation results. The resolution is 41×41 per colormap. We can notice that the optimal point intersects the contours for $\phi \approx \pi$ and $t_x/t_y \approx 1$, and overlaps the points for minimal CoPE and maximal η_D .

Width [nm]	Period [nm]	Φ / π	T_x	T_y	T_x/T_y	CoPE	CrPE	Eta_D [%]	Note	
120	200	0.9731	0.2035	0.4922	0.4135	0.021	0.1208	85.1864	Paper	
160	174	1.0158	0.3121	0.5635	0.5538	0.0159	0.1916	92.3312	Max CrPE	
136	230	0.9975	0.207	0.2126	0.9739	8.40E-06	0.044	99.9809	Min CoPE	
138	230	1.0098	0.2181	0.2042	1.0679	5.88E-05	0.0446	99.8684	Min CoPE	
140	230	1.019	0.2272	0.1955	1.162	2.90E-04	0.0446	99.3535	Interest	

Fig. 19. Points of interest in numerical simulation results.

To compare the performance of our chosen design for the nano-antenna array, we conducted a numerical simulation of the beam deflector blazed grating in Ansys's Lumerical program. The z-axis distance was 1-meter from the meta-surface. The DGMOE was excited with three different excitation polarizations: RHCP, LHCP and Linear. We took ten replicas of the eight-level discretized blazed grating with 136-nm width and 230-nm period, resulting in a blazed grating period of $\Lambda \approx 3.68\mu\text{m}$.

The results for the three simulations were two $17,000 \times 17,000$ matrices - E_L and E_R , showing the electric fields of the LHCP and RHCP polarizations; and two 1×17000 vectors u_x and u_y , signifying the far-field dimensions for the x and y axes. Loading this data to my computer's RAM was not possible, since the data is formatted in Float64 format (requires 8-bytes per entry), resulting in the need for around 50GB of RAM to handle it. To overcome this issue, I've downsampled the data by a scale of 10, lowering the resolution to just $\sim \lambda/4$.

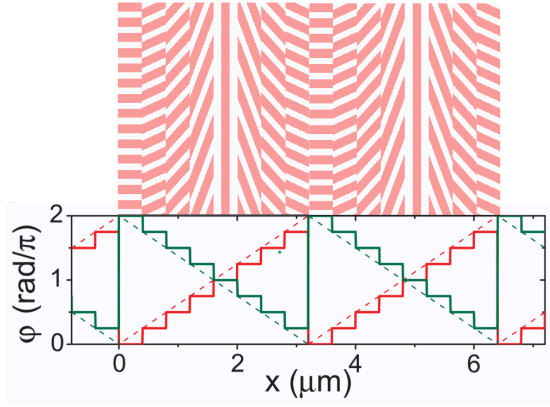


Fig. 20. Blazed grating with linear phase profile - eight levels of phase discretization plotted as steps.

To calculate the transmission efficiency, we need to perform an integration over the intensity of each spot, to calculate the power of each diffraction order P_{-1}, P_0, P_{+1} and then find the ratio between the total transferred power and the incident power. In addition, to calculate the deflection efficiency, we need to

calculate the ratio of $P_{-1} + P_{+1}$ to the total power transmitted. Both of these calculations require an integration, which was not possible to be conducted due to the hardware limitations of my computer.

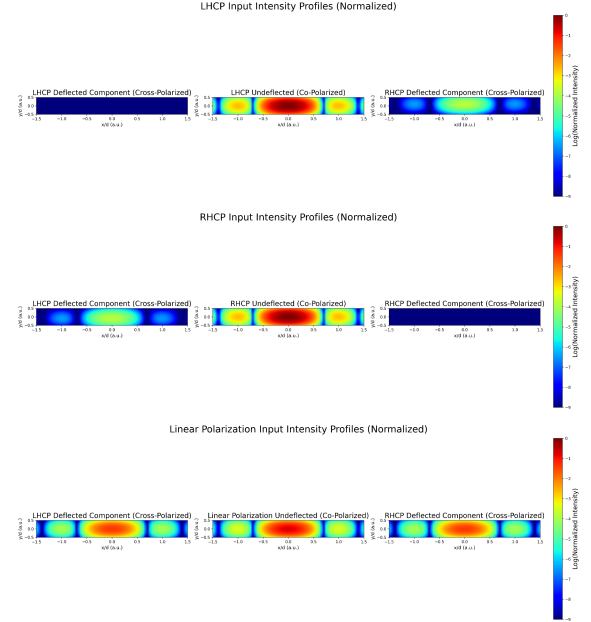


Fig. 21. Blazed grating DGMOE numerical simulation results, plotted in logarithmic scale, at 550-nm wavelength. From top to bottom - LHCP, RHCP, and linear polarized illumination results. Left to right - cross-pol LHCP, co-pol light, cross-pol RHCP. The intensity was normalized using the maximum intensity of the co-polarized light (center spot, $x=y=0$). The x-axis and y-axis are normalized by d - the distance between the spots. Logarithmic color scheme.

The results are not very promising - despite imitating the behavior of a half-wave plate pretty accurately, the deflection is not significantly better than seen in the research paper, evident by the fact that the middle spot (non-deflected) still demonstrates the dominant intensity. Although, the results closely resemble the behavior expected from the DGMOE as seen in the paper, with the linear polarization splitting into both RHCP and LHCP light, and the RHCP and LHCP switching up depending on which is the input and which is the output.

The high intensity of the co-polarized light leads us to conclude that the materials used for the DGMOEs are not ideal, due to their absorption and reflection properties. In fact, it

is not worthwhile to pursue further optimization, due to the fact that the CrPE in general is too low for these materials.

The crucial finding from these simulations is the fundamental trade-off between minimizing the CoPE and maximizing the CrPE. An optimization focused on minimizing CoPE results in near-perfect half-wave plate behavior, but at the cost of lower cross-polarized (deflected) intensity. Conversely, maximizing the CrPE increases the power of the deflected beams, but degrades device performance by allowing for higher co-polarized leakage into the undeflected zeroth-order spot.

7. Conclusion

This term paper has provided an in-depth review and analysis of the work by Lin et al., which presented a significant advancement in the field of flat optics. The authors successfully demonstrated that dielectric gradient metasurfaces, based on silicon nano-beam antennas, can overcome the high ohmic losses of previous metallic designs and achieve high efficiency in transmission mode. The operation of these devices is governed by the Pancharatnam-Berry geometric phase, which allows for precise, space-variant control of a light beam's wavefront.

The reviewed article successfully realized three fundamental optical components on an ultra-thin (100-nm) platform: an axicon, a beam deflector, and a lens. As verified by the theoretical reproductions in this paper, the axicon generated a high-quality non-diffracting Bessel beam, the beam deflector exhibited decent polarization-controlled steering with high efficiency, and the metalens achieved near-diffraction-limited focusing.

However, a critical analysis of the device's building block - the nano-beam waveplate - revealed that its performance was non-ideal at the 550-nm design wavelength due to unequal transmission coefficients for orthogonal polarizations. Further investigation suggested that the optimal operating point for polarization conversion was closer to 520-nm. Additionally,

the analysis confirmed that while the diffraction efficiency of the transmitted light was high, significant energy losses of up to 86% occurred due to material absorption and surface reflection. A better beam deflector is achieved at a different design width of 136-nm for the same duty-cycle, with a design principle of CoPE minimization. This improvement, however, introduces a critical trade-off, as it lowers the CrPE which corresponds to higher absorption losses within the material.

Despite these limitations, the work by Lin et al. stands as a successful proof-of-concept, paving the way for the development of low-cost, lightweight, and compact optical elements that work in transmission mode by using dielectric materials. If we change the materials to have better reflective and absorption properties, or work at a different wavelength, this is a very promising aspect for the future of nanophotonics.

Future work could also explore entirely different nano-antenna geometries to overcome the specific CoPE-CrPE trade-off inherent in this rectangular nano-beam design

8. Acknowledgements

I would like to express my greatest gratitude to Raghav for his invaluable assistance in this paper, helping me with the numerical meta-atoms and device simulations by developing the frameworks and running the simulations. Without his help, I would not have been able to complete this task, especially not Chapter 6 of this term paper, which provides important insights to the analysis of the research paper.